

TerraScan: AI Solution for Low Cost Agricultural Soil Mapping

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Table of contents



01

Background Research

Details of the current scenario in the world

02

The Problem

The issue with the current process

03

Proposed Solution

An idea on how to address the problem

04

Design

Plan a framework for the solution

05

Implement

Build and iterate through the design

06

Results

Analyze the output of the solution



01

Background Research



The Discovery Process

Outline

Abstract

Keywords

1. Introduction

2. Methodology

3. Result

4. Discussion

5. Future direction and conclusion

References

Show full outline

Cited by (184)

Figures (2)

Get citation

Abstract

Soil is the basis of production in agriculture activities. The combination of intensive farming activities, improper farming practices, rainfall regimes, and topography conditions that taken place in agricultural land lead to the soil erosion problem. Soil erosion is the major constraint to agriculture that affects the yield production and degraded environmental sustainability. Furthermore, soil erosion that occurs in the agricultural area has jeopardized the sustainability of agriculture activities. Asia is one of the major agricultural producers in the world. It is essential to know how to mitigate soil erosion in Asian agricultural land. This systematic review aims to analyze the existing literature on research that has been done on control practices that had been taken in Asia agricultural land towards soil erosion. This article is guided by the PRISMA Statement (Preferred Reporting Items for Systematic reviews and Meta-Analysis) review method. The authors systematically reviewed the literature to study the control practices that been taken and tested to control soil erosion on the agricultural land in Asia. Accordingly, this systematic review identified 39 related studies about the topic based on the Web of Science and Scopus databases. This article divided the control practices into three main themes, which are agronomic practices, agricultural practices, and mechanical practices. The three main themes then produced a total of 11 sub-themes. Further specific and sustained research is needed to tackle this severe environmental problem through a better method, such as this systematic review method. The systematic review helps farmers and policymakers to implement the most practical approach to control and reduce soil erosion.

Previous article in issue

Next article in issue

FEEDBACK



Modeling Approaches to Assess Soil Erosion by Water at the Field Scale with Special Emphasis on Heterogeneity of Soils and Crops

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Abstract: Information on soil erosion and related sedimentation processes are very important for natural resource management and sustainable farming. Plenty of models are available for studying soil erosion but only a few are suitable for dynamic soil erosion assessments at the field-scale. To date, there are no field-scale dynamic models available considering complex agricultural systems for the simulation of soil erosion. We conducted a review of 15 different models evaluated based on their representation of the processes of soil erosion by water. Secondly, we consider their suitability for assessing soil erosion for more complex field designs, such as patch cropping, strip cropping and agroforestry (alley-cropping systems) and other land management practices. Several models allow daily soil erosion assessments at the sub-field scale, such as EPIC, PERFECT, QUEST, EP4, TCIP, SLEMSA, APSIM, RUSLE, WaNuCAS, SCIAF, and CERIAS. However, further model development is needed with respect to the interaction of components, i.e., rainfall intensity, overland flow, crop cover, and their scaling limitations. A particular shortcoming of most of the existing field scale models is their one-dimensional nature. We further suggest that platforms with modular structures, such as SIMULACT and APSIM, offer the possibility to integrate soil erosion as a separate module/component and link to GIS capabilities, and are more flexible to simulate fluxes of matter in the 2D/3D dimensions. Since models operating at daily scales often do not consider a horizontal transfer of matter, such modeling platforms can link erosion components with other environmental components to provide robust estimations of the three-dimensional fluxes and sedimentation processes occurring during soil erosion events.

Keywords: erosion and sedimentation processes; model categorization; complex cropping systems; governing equations; application

1. Introduction

Check for updates

Citation: Raza, A.; Ahrends, H.; Habib-Ul-Rahman, M.; Gaiser, T. Modeling Approaches to Assess Soil Erosion by Water at the Field Scale with Special Emphasis on Heterogeneity of Soils and Crops. *Land* **2023**, *12*, 422. <https://doi.org/10.3390/land12040422>

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Perspective

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Deep Vertical Rotary Tillage: A Sustainable Agricultural Practice to Improve Soil Quality and Crop Yields in China

by Wenlong Zhang ¹ , Jinhua Shao ^{1,2} , Kai Huang ², Limin Chen ³, Guanghui Niu ³, Benhui Wei ⁴ and Guoqin Huang ^{1,*}

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(This article belongs to the Special Issue Soil Health and Crop Management in Conservation Agriculture)

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Review Reports

Versions Notes

Abstract

Deep vertical rotary tillage (DVRT) is an innovative soil tillage technology that has been widely adopted in China and shown significant potential in enhancing soil quality, optimizing water use efficiency, and increasing crop yields across diverse ecological and agronomic conditions. DVRT utilizes a vertical spiral drill bit for deep plowing, which preserves soil structure, reduces soil compaction, and improves water retention, making it particularly effective in regions facing climatic challenges such as drought. This review synthesizes the effects of DVRT on soil's physical and chemical properties, crop root systems, photosynthesis, and water use efficiency, demonstrating its advantages in promoting robust root development and improving nutrient utilization. Although the technology has been applied successfully across various crops and regions, its nationwide adoption remains limited. This paper emphasizes the

Back

CATENA

Volume 193, October 2020, 104604

GIS-based soil erosion modelling under various steep-slope vineyard practices

Anton Pijl ^{1,*} , Lara E.H. Reuter ^b, Edoardo Quarella ^a, Teun A. Vogel ^b, Paolo Tarolli ^a

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<https://doi.org/10.1016/j.catena.2020.104604>

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Highlights

- Three vineyard practices were evaluated by spatial erosion models RUSLE and SIMWE.
- Terraces prevent widespread gradual erosion, in contrast with vertical cultivation.
- Intense rainstorms cause runoff concentration and erosion patterns



The Discovery Process - Data Points



ECONOMIC IMPACT OF EROSION

-  **€1.25 Billion/Year:** Total economic cost of severe soil erosion across 12 million hectares in the EU.
-  **0.24% Price Drop:** Reduction in agricultural land value for every 1% of topsoil lost.
-  **\$1,130 per Farm:** Average financial loss per farm in Uruguay due to initial topsoil degradation.



ENVIRONMENTAL & SEDIMENT ISSUES

-  **159 Tons of Sediment:** Volume of runoff prevented from entering waterways by a single field's catch basin.
-  **Economic Feedback Loop:** Declining land values and high upfront costs create a barrier to VRT adoption.



AGRICULTURAL LOSSES & RATES

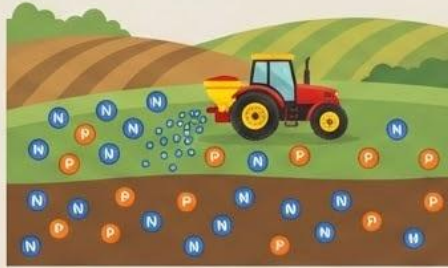
-  **0.43% Productivity Loss:** Average annual crop yield reduction caused by erosion in the European Union.
-  **19 tons/ha/year:** Peak erosion rates consistently found on agricultural lands.
-  **18.83 tons/ha/year:** Sustained "alarming" erosion rate even with sediment control.
-  **Targeted Hotspots:** Erosion concentrated on slopes and wheel tracks, ideal for VRT.

RESEARCH LIMITATIONS



-  **<3 Year Studies:** Most soil research is short-term, limiting the long-term data needed for effective policy.

Synthetic Over-saturation



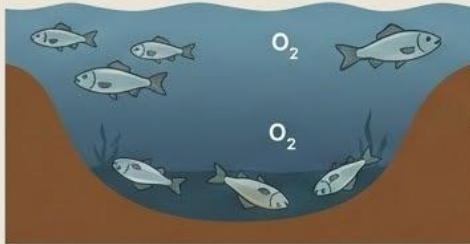
The VRT Gap: Inefficient fertilizer application due to lack of precision mapping.

Hydrological Transport (Runoff)



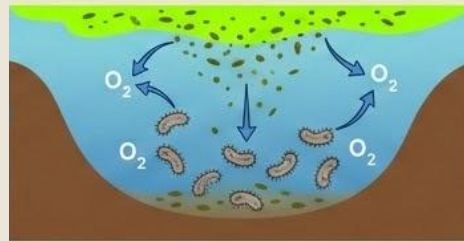
Leaching and Surface Runoff.

Benthic Hypoxia (The Dead Zone)

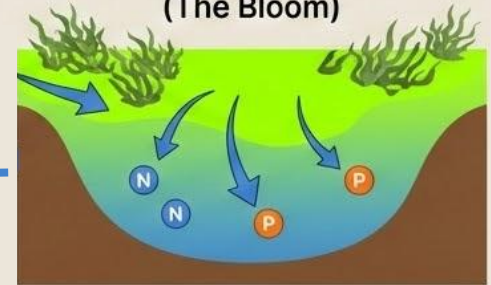


Hypoxia: Dissolved Oxygen levels drop below 2 mg/L, creating a biological 'Dead Zone'.

Microbial Respiration (Oxygen Consumption)



Hyper-Eutrophication (The Bloom)



Algal Bloom: Explosive biomass growth driven by nutrient stoichiometry.

02

The Problem



\$120,000

Small-scale farmers lack the budget for precision nutrient mapping

Source: <https://ageconsearch.umn.edu/>





Reclaiming the Farmer's Most Valuable Asset—Time

Pennsylvania Ag E&S(Erosion & Sediment) Plans

SECTION 2: SOIL LOSS

Major/Dominant Soil Types on Farm	T Value (tons soil loss/acre/year)

SOIL LOSS CALCULATIONS

Predicted Average Annual Soil Loss Based on Planned Crop Rotation and Management:

Field	Rotation Year(s)	Management(s) – Crop Rotations and Tilling/ Planting Methods	Predicted Average Annual Soil Loss	Soil Type T Value

State law requires documented nutrient levels to prevent runoffs

Requires farmers to **manually** measure and keep track of nutrient levels

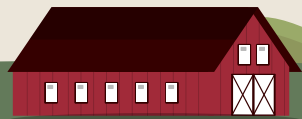
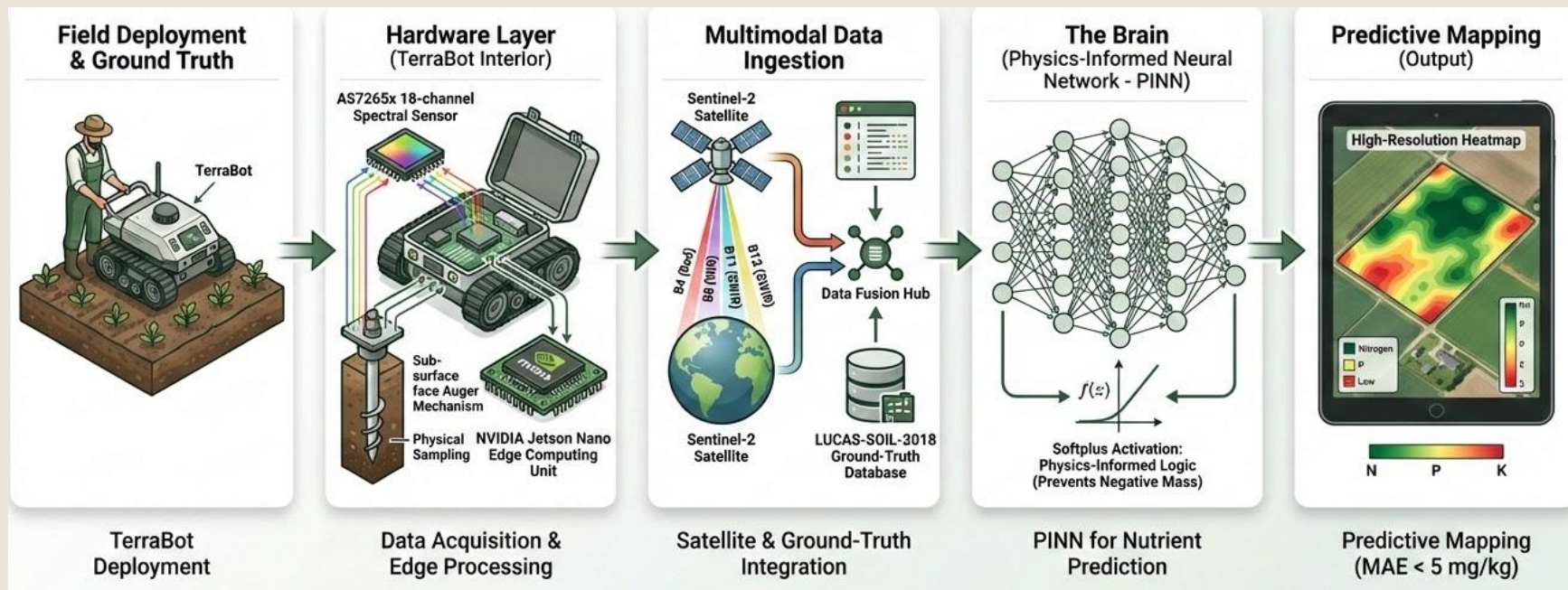
TerraScan provides **precision ground truth** to fill out these document **accurately**, helping the farmer also **save time**

03

Proposed Solution



Visual Abstract



The Hypothesis

If we apply **Physics-Informed Deep Learning** to Sentinel-2 satellite data, we can predict **Nitrogen**, **Phosphorus**, and **Potassium** levels within a scientifically viable margin of error ($< 5 \text{ mg/kg}$).



04

Design



TerraScan Technical Pipeline



Data Ingestion

API: Google Earth Engine (GEE)
Source: Sentinel-2 Harmonized MSI
Features: 12 Spectral Bands (B1-B12)
Scale: 20m Spatial Resolution



Ground Truth Engineering

Source: LUCAS-SOIL 2018 (628 Validated Samples)
Action: 99th Percentile Quantile Filtration
Process: Null-value remediation for Nitrogen (N), Phosphorus (P), and Potassium (K).



Feature Scaling

Method: StandardScaler (Mean Removal & Variance Scaling)
Framework: PyTorch Tensor Conversion
Split: 80/20 Randomized Train-Test Distribution



Physics-Informed Deep Learning

Model: 4-Layer MLP with Batch Normalization
Constraint: Softplus Activation (Ensures Positive-Only Output)
Optimizer: AdamW with Dynamic Learning Rate Scheduler

05

Implement



Physics-Informed Neural Network(PINN)

```
class TerraScanAI(nn.Module):
    def __init__(self):
        super(TerraScanAI, self).__init__()

        self.encoder = nn.Sequential(
            nn.Linear(12, 128),
            nn.LeakyReLU(0.1),
            nn.BatchNorm1d(128),
            nn.Dropout(0.25),

            nn.Linear(128, 256),
            nn.LeakyReLU(0.1),
            nn.BatchNorm1d(256),
            nn.Dropout(0.25),

            nn.Linear(256, 128),
            nn.LeakyReLU(0.1),
        )

        self.head = nn.Linear(128, 3)
        self.constraint = nn.Softplus()

    def forward(self, x):
        feat = self.encoder(x)
        raw_out = self.head(feat)
        return self.constraint(raw_out)
```

Depth Factor

I used the Softplus activation function because in the real world, soil chemicals cannot be negative. This **informs** the AI about the laws of **physics**

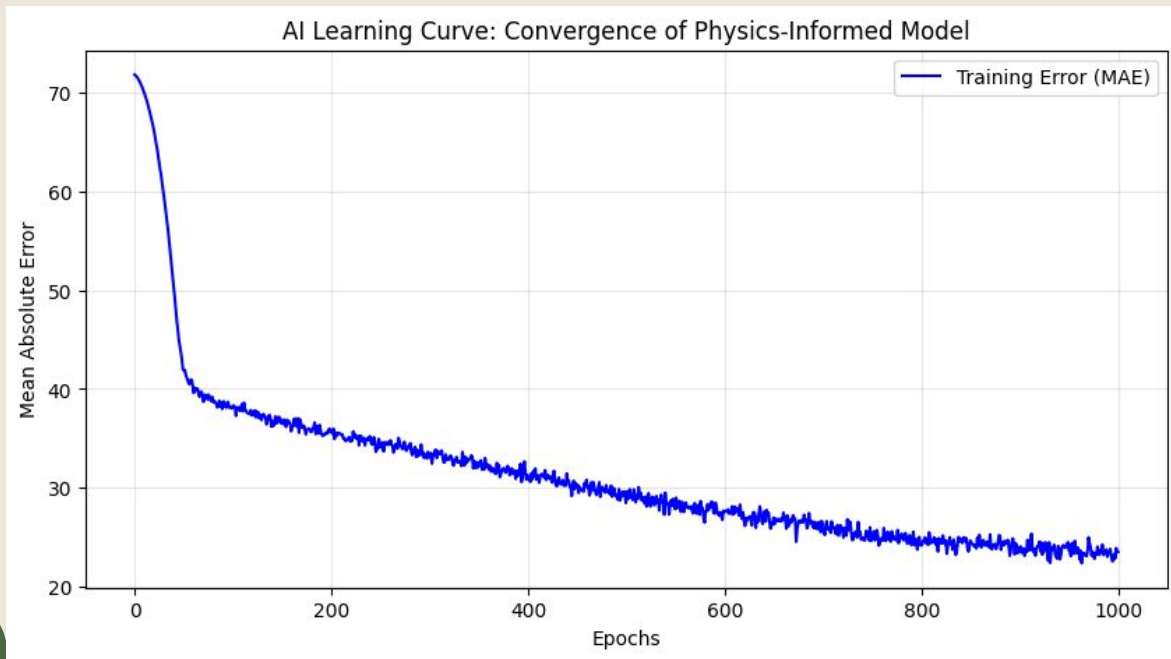
$$\text{Output} = \text{Softplus}(W \cdot x + b)$$

06

Results



Training Results



Training Process

Epochs: 1000

Valid Data Used: 628

Start MAE Loss: 71.8143

Final MAE Loss: 24.0726

Note: MAE drops sharply and stabilizes

Training Results

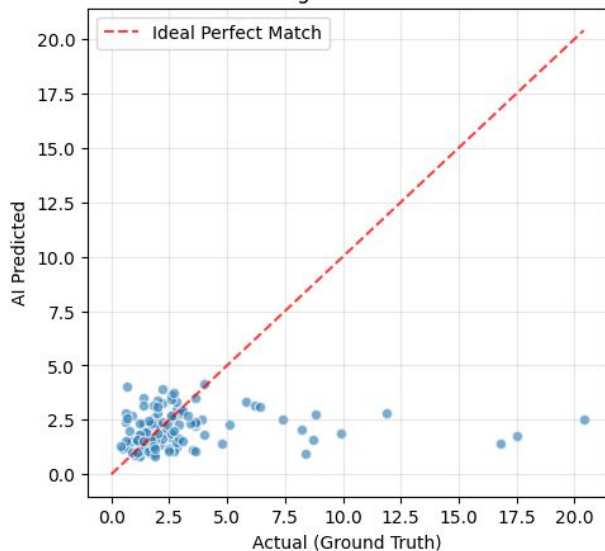


Optimizer: Used **AdamW** with a dynamic **learning rate** scheduler.

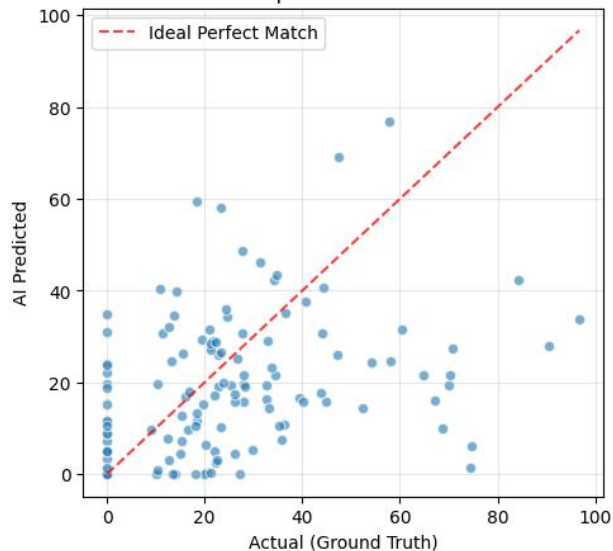
```
optimizer = optim.AdamW(model.parameters(), lr=LEARNING_RATE, weight_decay=1e-4)  
scheduler = optim.lr_scheduler.ReduceLROnPlateau(optimizer, 'min', patience=50, factor=0.5)
```


Testing Results

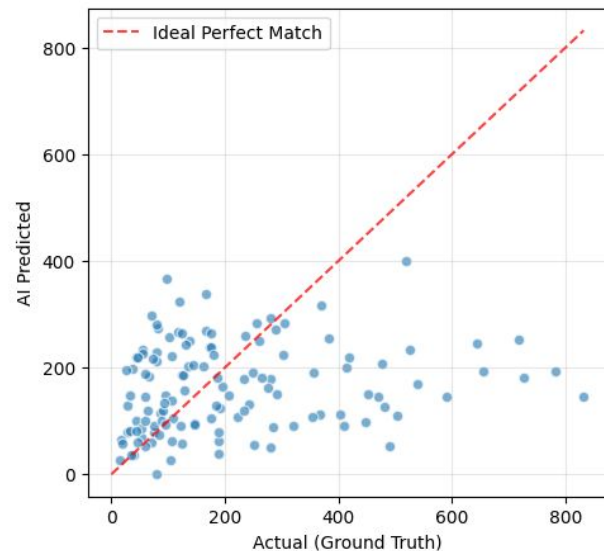
Nitrogen Prediction



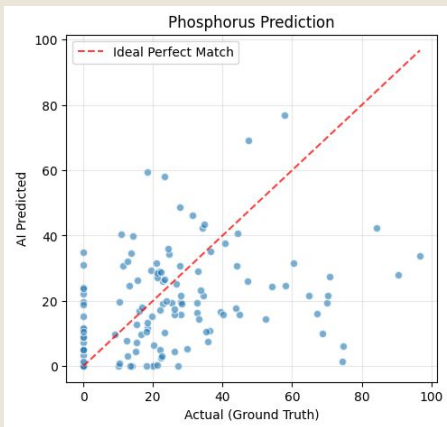
Phosphorus Prediction



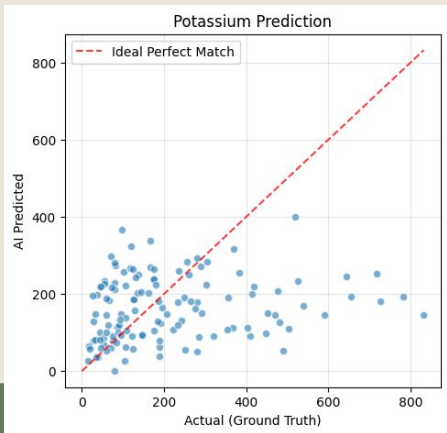
Potassium Prediction



Challenges

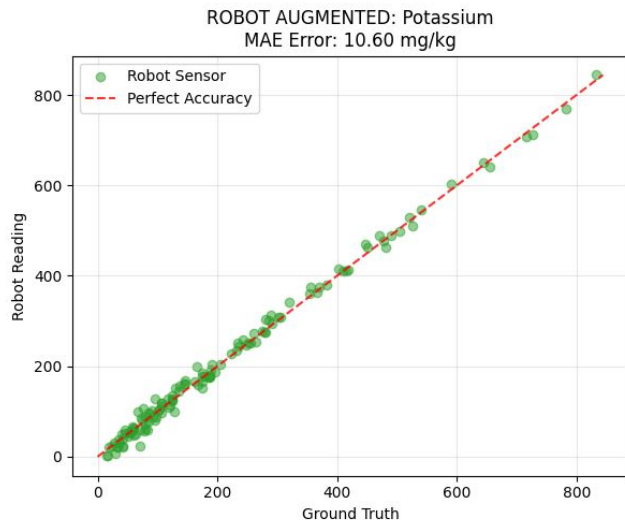
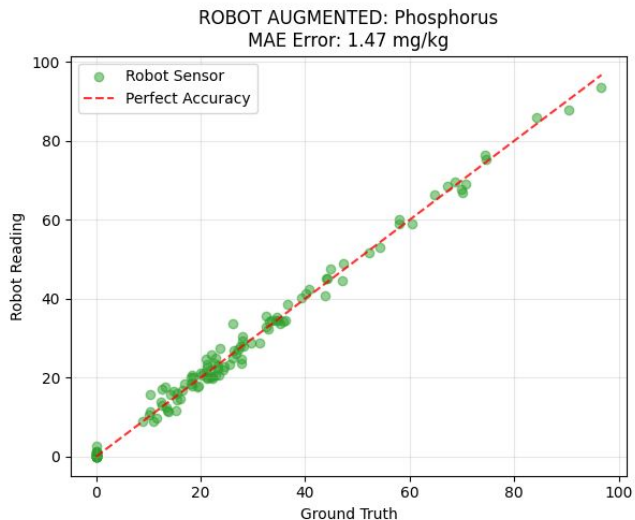
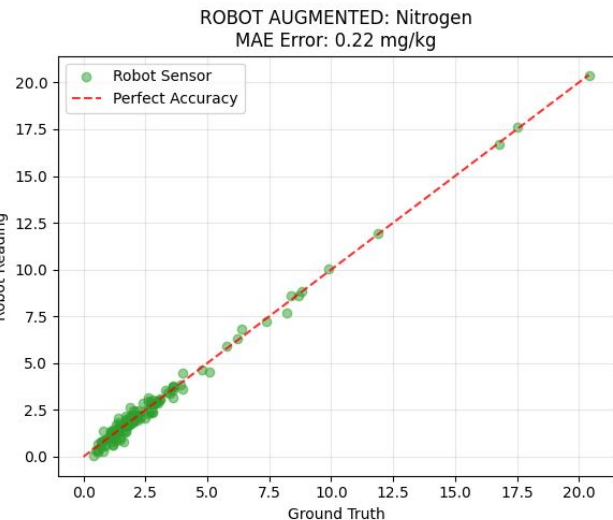


Phosphorus (with a Mean Absolute Error of 16.88) and Potassium (with a Mean Absolute Error of 135.95) are harder to identify and predict over soil moisture and texture in a single piece of data.

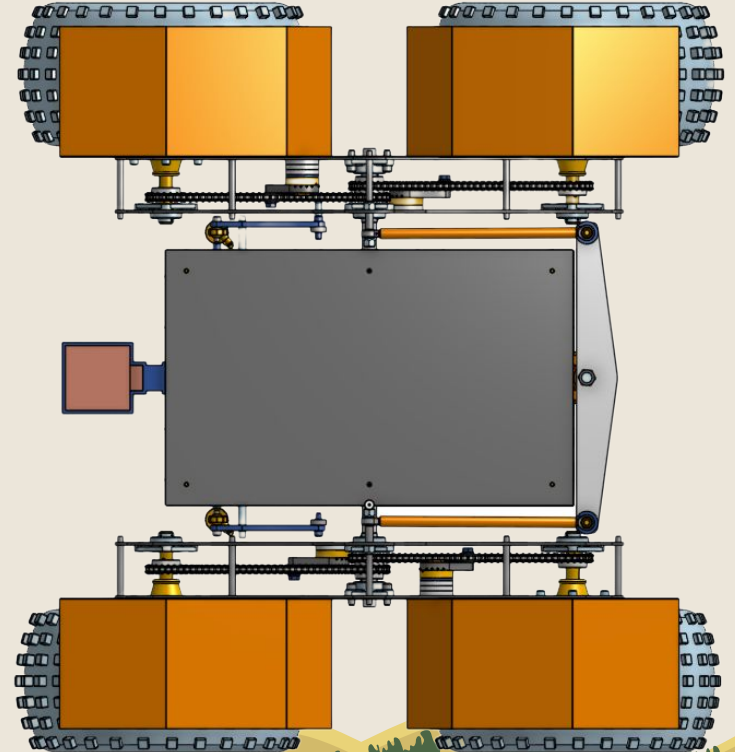
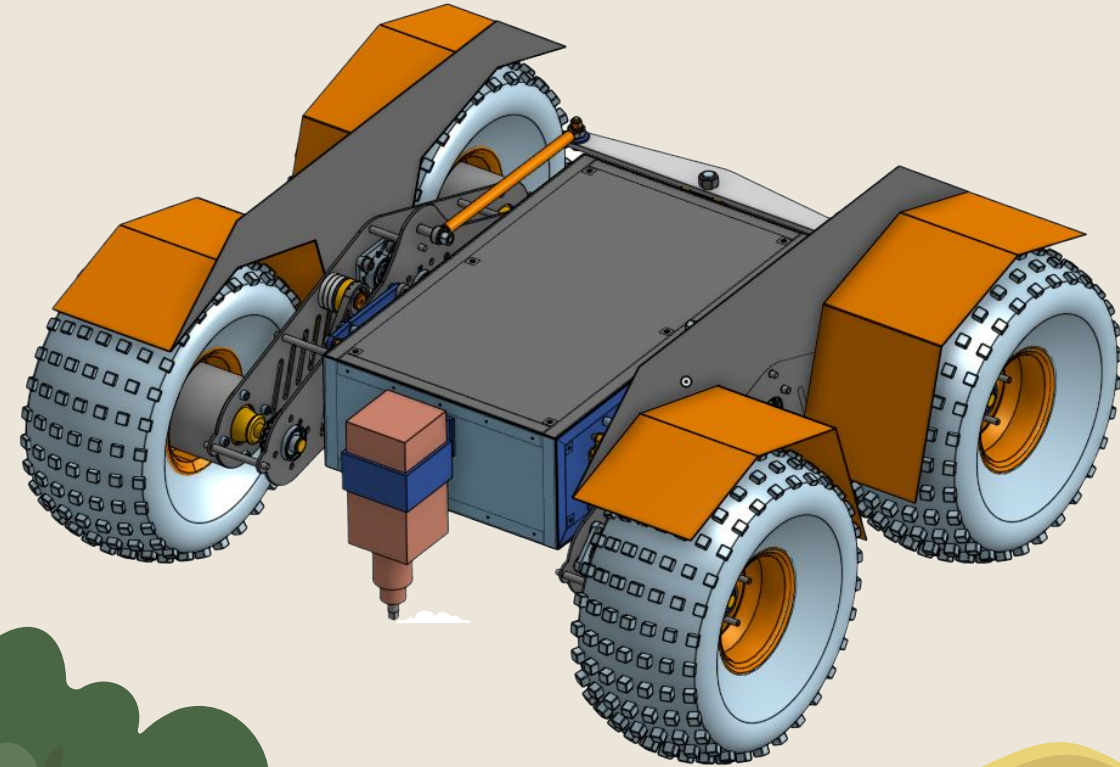


This circles back and necessitates the mix of both the physics-informed neural network and a physical robot that can pick up samples

Simulated Robot Testing Results



Robot Design



Robot Design

The Brain: "NVIDIA Jetson Nano" or "Raspberry Pi 5 (Edge Computing Unit)."

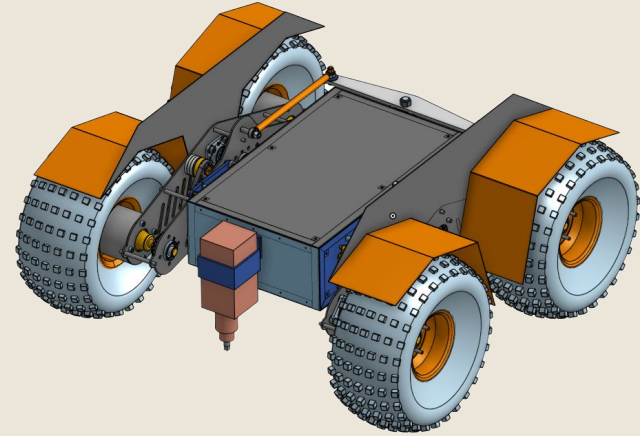
The Eye: "AS7265x 18-Channel Spectral Sensor" or "NIR Spectrometer Module."

The Connectivity: "LoRaWAN Transceiver" (for long-range farm data).

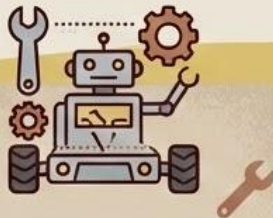
The Power: "12V LiFePO4 Battery Bank with Solar Controller."

The Movement: "High-Torque DC Geared Motors with Encoder Feedback."

Estimated Total Cost: \$485.00



Further Steps



1. BUILD TERRABOT

Physical robot construction based on proven hypothesis.



2. ACADEMIC PARTNERSHIPS

Carnegie Mellon University

Carnegie Mellon University (CMU)

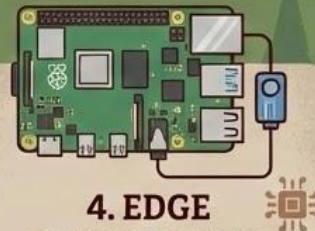


Penn State University



3. MODEL ENHANCEMENT

Increase data size and epochs for improved accuracy.



4. EDGE INTEGRATION

Integrate model onto Raspberry PI or other Edge device.

Thanks!

Do you have any questions?

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